A Reconfigurable Digital Compute-in-Memory Architecture With 2:4 Compressed Weight Storage for Highly Sparse Fusion Neural Network

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***Abstract*— The fusion of convolutional neural network (CNN) and spiking neural network (SNN) harnesses their complementary strengths in edge video tasks. However, ongoing challenges remain in the low compatibility between two neural networks and the shortage of storage resources. In this brief, we propose a reconfigurable digital compute-in-memory (RDCIM) architecture featuring 2:4 structured weight pruning, tailored for the efficient fusion of CNN and SNN. Specifically, differential frame convolution (DFC) is adopted in place of conventional convolution. Therefore, the RDCIM fusion accelerator can operate in either DFC mode or SNN mode based on the sparsity of activations. First of all, fine-grained 2:4 structured sparsity enables compressed weight storage, thereby doubling the throughput. Secondly, a novel bitwise in-memory Booth multiplier is implemented to support both 4-bit and 8-bit activations, offering flexible computation. Thirdly, the high sparsity inherent in both differential frames and input spikes allows for the concurrent support of three sparse optimization strategies: coarse-grained zero skipping, medium-grained bank gating, and fine-grained precharge gating. Implemented by TSMC 28nm technology, our proposed architecture achieves the peak energy efficiency of 42.02 TOPS/W for DFC and 0.02 pJ/SOP for SNN.**

***Index Terms*— Compressed weight storage, digital compute-in-memory (DCIM), fusion neural network, sparse optimization strategy.**

I. Introduction

RECENTLY, the fusion neural network, namely the integration of convolutional neural network (CNN) and spiking neural network (SNN), has seen broad application in areas ranging from image classification, object detection to target tracking [1], [2], [3], [4]. This is attributed to the fact that the fusion neural network can deliver CNN's robust feature extraction alongside SNN's brain-like capability in processing sparse and temporal data. Consequently, such hybrid systems achieve an optimal balance among high accuracy, low power consumption, and reduced inference delay, offering significant advantages for edge deployments, such as in application-specific integrated circuit (ASIC) [5], [6], [7], [8].

As shown in Fig. 1(a) and Fig. 1(b), conventional CNN and SNN are characterized by low network compatibility. Therefore, most hybrid accelerators, such as C-DNN [2] and the dual in-memory chip [3], utilize heterogeneous architectures to process CNN and SNN separately, leading to minimal resource reuse. To overcome this limitation, we explore computational paradigms that bridge CNN and SNN characteristics.

For edge video tasks, differential frame convolution (DFC) outperforms conventional CNN by leveraging inter-frame similarities to skip zero activations, exhibiting properties analogous to SNN [9], [10], [11], [12]. Specifically, DFC's computational paradigm aligns with SNN in three key aspects: 1) both networks demonstrate input sparsity of over 75% on average [10], [13]; 2) computations in both involve bitwise AND operations and accumulation operations; 3) the membrane potential accumulation in SNN is compatible with the accumulation storage mechanism introduced by DFC. These parallels inspire us to integrate DFC and SNN within hybrid accelerators.

In this brief, we propose a reconfigurable 8T-SRAM-based digital compute-in-memory (RDCIM) fusion accelerator with innovative bitwise Booth multiplication, notable for its 2:4 weight compression in compute-in-memory (CIM) storage and three sparse optimization strategies. Consequently, this architecture can efficiently support the diverse computational demands of both DFC and SNN. The key contributions of this brief are summarized as follows:

- To accelerate edge video applications, we introduce a RDCIM fusion accelerator that integrates DFC and SNN, leveraging their shared features of input sparsity, bitwise AND, and additional memory.
- For the first time, we apply the 2:4 compressed weight storage pattern with 3-bit indices to the CIM architecture, achieving over 40% reduction in storage resources without accuracy degradation.
- A novel in-memory Booth multiplier is designed to facilitate efficient fusion bitwise computing with simplified Booth encoding, supporting both 4-bit and 8-bit activation precision under DFC mode.
- We utilize multi-grained sparse strategies: zero skipping, bank gating, and precharge gating. The single bank of the RDCIM macro can be disabled by bank gating, and the dynamic logic circuit for 8-bit storage can be deactivated by precharge gating.

II. Background and Model

A. Differential Frame Convolution

As illustrated in Fig. 1, unlike the conventional CNN, DFC initially generates the highly sparse feature map $\Delta x(t_1)$

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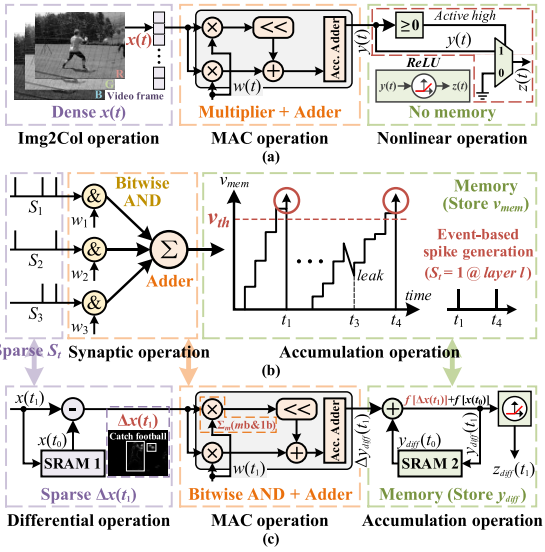


Fig. 1. Schematic diagram of (a) conventional CNN, (b) SNN, and (c) DFC. Both SNN and DFC share similar characteristics, including input sparsity, bitwise AND and accumulation operations, and additional memory requirement. MAC denotes multiply-accumulate.

through inter-frame differencing between $x(t_1)$ and $x(t_0)$, then performs sparse convolution f, and ultimately reconstructs the single-frame image $y_{diff}(t_1)$ linearly via accumulation operations [14]. The entire image processing can be expressed as

$$\begin{cases} \Delta x(t_1) = x(t_1) - x(t_0) \\ y_{diff}(t_1) = f\left[\Delta x(t_1)\right] + f\left[x(t_0)\right] \\ z_{diff}(t_1) = g\left[y_{diff}(t_1)\right], \end{cases} \tag{1}$$

where $z_{diff}(t_1)$ is generated by applying the non-linear activation function g to the reconstructed $y_{diff}(t_1)$ at time t_1.

B. Spiking Neural Network

Spiking neurons, modeled after the leaky integrate-and-fire (LIF) mechanism, share similar operational characteristics with DFC, including the integration of synaptic inputs and the accumulation of membrane potential. Additionally, neuronal computations also involve the leakage of membrane potential and the generation of output spikes [15]. Therefore, the neuron dynamics can be formulated as

$$\begin{cases} u_l^t[i] = \sum_j w_l[i,j] \times s_{l-1}^t[j] \\ v_l^t[i] = a \times v_l^{t-1}[i] \times \left(1 - s_l^{t-1}[i]\right) + u_l^t[i] \\ s_l^t[i] = H\left(v_l^t[i] - v_{th}\right), \end{cases} \tag{2}$$

where $u_l^t[i]$, $v_l^t[i]$, and $s_l^t[i]$ correspond to the input current, membrane potential, and spike of neuron i in layer l at time step t. Also, w_l is the weight in layer l, and a is the leaky factor. The function $H(x) = 0$ if $x < 0$; otherwise, $H(x) = 1$.

C. Fine-Grained 2:4 Structured Pruning

A fine-grained 2:4 structured sparsity technique, originally pioneered by NVIDIA, prunes 2 out of every 4 weights, yielding 6 different sparsity patterns [16]. This implementation necessitates storing additional indices per weight group to record the positions of the 2 unpruned elements. This pruning method can efficiently accelerate various neural networks while maintaining the same accuracy as the dense pattern. Notably, this compressed storage format achieves a reduction approaching 50% in memory capacity relative to the dense storage format, despite the overhead of additional indices.

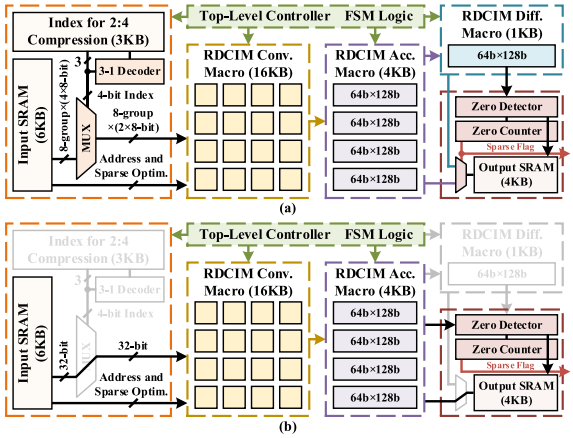


Fig. 2. Overall architecture of RDCIM fusion accelerator configured for (a) DFC mode, and (b) SNN mode. Optim., Conv., Acc., and Diff. represent optimization, convolution, accumulation, and differencing, respectively.

III. Architecture Design

A. Overall Architecture

Fig. 2 illustrates the overall architecture of the RDCIM fusion accelerator, highlighting its capability for reconfigurable processing of DFC and SNN. With a total capacity of 34KB, the fusion accelerator consists of three primary components: input/output SRAM (10KB), index storage for 2:4 compression (3KB), and 21 RDCIM macros for linear operations (21KB). Depending on the sparsity of activations, 21 RDCIM macros can be configured for either DFC mode or CNN mode. In DFC model, the input data undergoes sparse convolution, accumulation, and differencing operations. If the final results approach a dense state, the fusion accelerator will output non-differenced results; otherwise, it will remain in DFC mode.

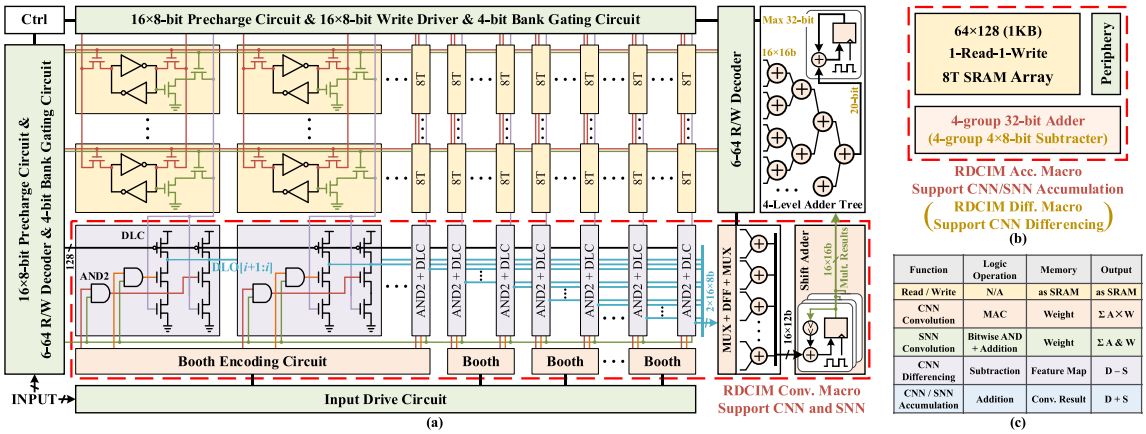
B. 2:4 Compressed Weight Storge

As depicted in Fig. 3(a), the single RDCIM macro incorporates a 64 × 128.8 T-SRAM array, where each row can accommodate 16 8-bit weights. Through the 2:4 compressed weight storage, the storage capacity of each row is equivalent to holding 32 8-bit weights, doubling the throughput. To enhance the compression ratio, 3-bit indices are utilized instead of the previous 4-bit indices to select the corresponding activations. Specifically, the fourth 1-bit index is decoded by 3-bit storage indices in Fig. 2(a), saving 1KB of storage space. In total, the 2:4 compressed weight storage pattern with 3-bit indices can achieve a reduction of over 40% in CIM storage without any loss of accuracy.

C. RDCIM Macro and Data Flow

The RDCIM macro can be divided into the RDCIM conv. macro and the RDCIM non-conv. macro. As shown in Fig. 3(a), the RDCIM conv. macro comprises a

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Function	Logic Operation	Memory	Output
Read / Write	N/A	as SRAM	as SRAM
CNN Convolution	MAC	Weight	Σ A×W
SNN Convolution	Bitwise AND + Addition	Weight	Σ A & W
CNN Differencing	Subtraction	Feature Map	D – S
CNN / SNN Accumulation	Addition	Conv. Result	D + S

Fig. 3. (a) Detailed description of RDCIM conv. macro, supporting both DFC and SNN. Within the red dashed box are 16 bitwise in-memory Booth multipliers. DLC and DLO stands for dynamic logic circuit/output. Mult. represents multiplication. (b) Brief description of RDCIM non-conv. macro configured for either acc. mode or diff. mode. (c) Table of various logic operations, in-memory storage, and outputs under different functions. A and W denote activation and weight, respectively.

64 × 128.8 T-SRAM array, input drive circuits, precharge circuits, read/write decoders, write drivers, bank gating circuits, bitwise in-memory Booth multipliers (within the red dashed box), adder trees, and shift adders, all of which cater to the diverse computational demands of DFC and SNN. Each conv. macro is partitioned into four 64×32 banks along the row, with each operating independently in either work mode or gating mode. Benefiting from 2:4 compressed weight storage, MAC operations on an 8 × 8 tile using a 3 × 3×3 kernel can be completed by a single conv. macro without updating weights.

Fig. 3(b) provides a brief description of the one-cycle RDCIM non-conv. macro, which supports mode switching between the acc. and diff. macro configurations. Fig. 3(c) lists the five types of logic operations, in-memory storage, and outputs of all RDCIM macros. Specifically, the RDCIM acc. macro stores the convolution result of a single frame, which is first read out, then added to the convolution result of the differential frame, and finally written back to overwrite the previous result. The RDCIM diff. macro stores the feature map of a single frame, which is read out and subsequently overwritten by the new feature map. Compared to dual-port SRAM, the non-conv. macro can eliminate external pipeline overhead and optimize storage-compute timing.

D. Bitwise In-Memory Booth Multiplication

The performance of the RDCIM fusion accelerator can be optimized by integrating the customized RDCIM architecture with the novel bitwise in-memory Booth multiplier. The radix-4 Booth algorithm is adopted to halve the number of partial sums (PPs), generating five types of PPs: $\{0, W, -W, 2W, -2W\}$, as illustrated in Fig. 4. To support both 4-bit and 8-bit activation under DFC mode, the Booth-multiplication-based RDCIM architecture flexibly utilizes one/two cycles of a 4b×8b multiplier to complete 4b×8b/8b×8b multiplications, respectively. In contrast, SNN mode performs two independent 1b×8b multiplications per cycle. Therefore, SNN mode achieves a 4× higher throughput than DFC mode by leveraging hardware parallelism for binary operations.

As presented in Fig. 5(a), the Booth encoder achieves a 25.9% area reduction compared to [17] by adding the extra 1'b1 to exploit overflow characteristics, generating the correct

Images			*Encoder*			*Intermediate Results*			*Target*
I[2*i*+1]	*I*[2*i*]	*I*[2*i*–1]	*Eql*	*Dul*	*Sgn*	*Dynamic Logic Output*	*MUX+DFF+ MUX Output*	*Ext "1"*	*Partial Sum*
0	0	0	0	1	0	{9{1'b1}}	9'b0	0	0
1	1	1	0	1	1		{9{1'b1}}	1	0
0	0	1	1	0	0	~{*W*[7], *W*}	{*W*[7], *W*}	0	*W*
0	1	0	1	0	0			0	*W*
1	0	1	1	0	1		~{*W*[7], *W*}	1	–*W*
1	1	0	1	0	1			1	–*W*
0	1	1	1	1	0	{~*W*, 1'b1}	{*W*, 1'b0}	0	2*W*
1	0	0	1	1	1		{~*W*, 1'b1}	1	–2*W*

Fig. 4. Truth table for novel bitwise in-memory Booth multiplication with simplified Booth encoding. *W* represents the weight of the neural network.

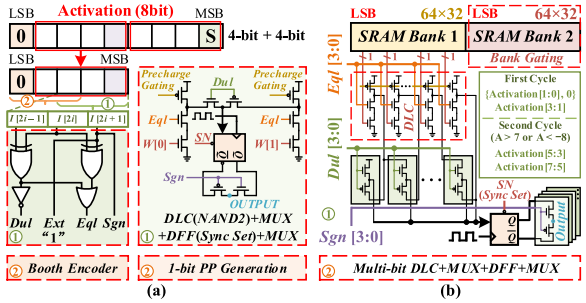


Fig. 5. (a) Schematic diagram of simplified Booth encoder and 1-bit PP generation circuit. (b) Illustration of multi-bit PPs generation circuit, consisting of DLC and MUX+DFF+MUX. The entire Booth multiplication is accomplished in two cycles.

PP of 9'b0. The two PPs generated by 4-bit activation necessitate two Booth encoders and two multi-bit PPs generation circuits (i.e., *DLC*+*MUX*+*DFF*+*MUX*) to operate in one cycle, as depicted in Fig. 5. In the DLC, the gates of two NMOS transistors are connected to *Eql* and *W* respectively, and $Eql = 0$ only when PP is 0. During the generation of PPs, Fig. 4 demonstrates that each 8-bit intermediate result is extended by an additional 1-bit. If the PP is $\pm W$ (i.e., $Dul = 0$), the most significant bit (MSB) of DLO is extended to the highest bit of the weight; otherwise, DLO is shifted left by 1-bit and the least significant bit (LSB) is set to 1'b1. Depending on the signal *Sgn*, the PPs are either buffer-based DFF results or inverter-based DFF results. Moreover, the *Ext* "1" is connected to the input *Cin* of the full adder or adder tree, simplifying the computation when $I\ [2i+1] = 1'b1$.

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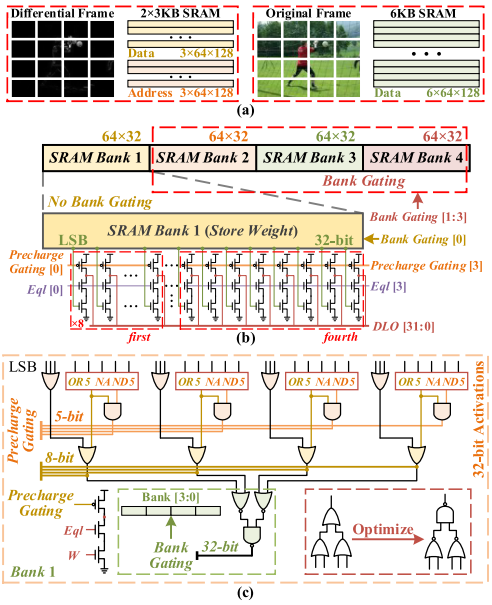


Fig. 6. (a) Schematic diagram of feature map storage for differential frames and original frames. (b) Illustration of medium-grained bank gating and fine-grained precharge gating. DLCs are only partially displayed. (c) Optimized zero detector for bank gating and precharge gating.

E. Three Sparse Optimization Strategies

Due to the highly sparse nature of both differential frames and input spikes, three sparse optimization strategies are applied to accelerate the RDCIM architecture: zero skipping, bank gating, and precharge gating. Fig. 6(a) presents a visual illustration of the feature map storage for differential frames and original frames, catering to the storage requirements of compressed storage and conventional storage, respectively. In the compressed storage format, the input SRAM stores both 16 non-zero channel-wise values and their corresponding addresses for convolution operations. Therefore, the coarse-grained zero skipping technique can skip $n\times128$-bit zero feature maps in a single cycle, where the maximum value of n is 64, matching the row count of SRAM.

As depicted in Fig. 6(b), medium-grained bank gating can disable multiple 64×32 SRAM banks within an entire 64×128 SRAM array. The bank-gated SRAM retains the stored information but lacks the ability to perform read and write operations. The 4-bit precharge gating signal controls 4 groups of 2×8 DLCs in a single SRAM bank. If one group of 2×8 DLCs is precharge-gated, the corresponding DFFs for in-memory Booth multiplication are also clock-gated. Therefore, the X-state generated by DLCs cannot be stored and propagated by these DFFs.

Fig. 6(c) provides a detailed description of the optimized zero detector, which has achieved a 21.7% reduction in area through logic optimization. For two-cycle in-memory Booth multiplication, the state of the first cycle is decided by the 8-bit precharge gating signal. If the first cycle is in work mode, the state of the second cycle is determined by the 5-bit precharge gating signal; otherwise, two cycles are in gating mode. Given that differential frames can be expressed using the lower 4 bits with a probability of 74.3% in the HARDVS dual-modality dataset [18], the second cycle primarily operates in gating mode, avoiding resource wastage and timing constraints in single-cycle multiplication.

TABLE I
ACCURACY RESULTS ON HARDVS, CIFAR-100, AND IMAGENET

		HARDVS	CIFAR-100	ImageNet
ResNet-18	CNN	53.19%	75.23%	69.89%
	SNN	52.12%	74.62%	65.07%
MobileNetV2	CNN	48.38%	68.24%	71.65%
	SNN	42.13%	59.81%	63.74%

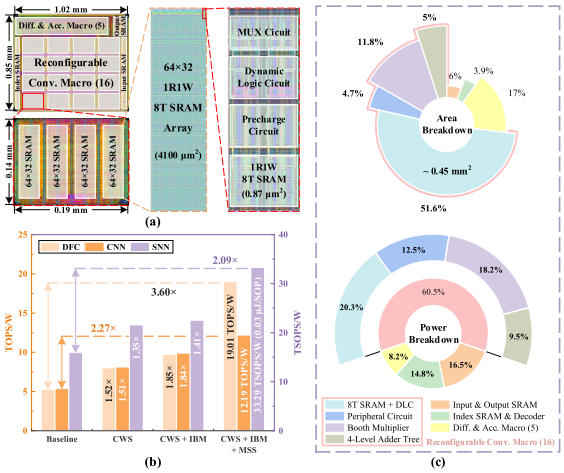


Fig. 7. (a) Layout photographs of RDCIM fusion accelerator. (b) Technique breakdown in energy efficiency. DFC and CNN modes are on the left coordinate axis with sparsity levels of 50% and 20%. (c) Area breakdown and power breakdown.

IV. IMPLEMENTATION AND RESULTS

A. Accuracy Results of 2:4 Structured Pruning

We validate the fine-grained 2:4 structured pruning method using quantized 8-bit weights based on the HARDVS dataset, which consists of video cutting frames [18], [19]. Specifically, the weights in a trained dense network are pruned to attain the 2:4 sparsity pattern. Although the pruning step leads to a significant drop in accuracy, the subsequent iterative retraining steps gradually restore it to the level of the unpruned network. As shown in Table I, the RDCIM architecture achieves a balanced accuracy for both CNN and SNN on the HARDVS, CIFAR-100, and ImageNet datasets, with each layer quantized to INT8.

B. Performance, Power, and Area (PPA) Analysis

Our proposed RDCIM fusion accelerator is designed using TSMC 28nm CMOS High Performance Compact Plus (HPC+) technology. With a total area of 0.87 mm^2, Fig. 7(a) presents a layout overview of the RDCIM fusion accelerator, highlighting the detailed layout of the custom-designed circuits. Fig. 7(c) further illustrates the area breakdown and power breakdown of the RDCIM fusion accelerator, respectively.

To evaluate the advantages of each technique, we refer to our prior work [20] as the baseline for technique breakdown analysis presented in Fig. 7(b). Compared with this baseline, the implementation of 2:4 compressed weight storage (CWS), novel in-memory Booth multiplication (IBM), and multi-grained sparse strategies (MSS) can boost energy efficiency by 1.52×, 1.31×, and 1.95× in DFC mode, respectively. Specifically, MSS represents the quintessential performance enhancement technique of our proposed

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TABLE II
COMPARISON WITH PREVIOUS WORKS

	JSSC'24 [2]	JSSC'22 [10]	ISCAS'24 [14]	This work
Node	28 nm	65 nm	28 nm	**28 nm**
Capacity	552 KB	196 KB	36 KB	**34 KB**
Architecture	Digital Circuit	Near-memory Digital Circuit	Digital Circuit	**8T SRAM Digital CIM**
Application	CNN & SNN	DFC	DFC	**DFC & SNN**
Activation Precision	4-16b@CNN 1-16b@SNN	4b/8b @CNN	4b/8b @CNN	**4b/8b@CNN 1b@SNN**
Weight Precision	4b/8b	8b	8b	**8b**
Dataset	CIFAR-100 ImageNet	63K Video Frames	N/A	**HARDVS CIFAR-100 ImageNet**
Accuracy [1]	75.9%@Res18 70.2%@Mob	N/A	N/A	**75.2%@Res18 71.7%@MobV2**
Pruning	No	No	No	**Yes**
Frequency (MHz)	50-200	25-100	200	**284**
Area Efficiency (TOPS/mm^2)	N/A	0.11 @CNN [2]	0.21 @CNN	**0.72 @CNN** [4]
Energy Efficiency (TOPS/W)	24.5 @CNN	0.129-13.3 @CNN	0.2-41.1 @CNN [3]	**9.75-42.02 @CNN** [4]
Energy Efficiency (pJ/SOP)	1.1 @SNN	N/A	N/A	**0.05-0.02 @SNN** [5]

[1] *Accuracy of ResNet-18 on CIFAR-100 and MobileNet series on ImageNet.*
[2] *Area efficiency = Energy efficiency × Max power / Chip area (slightly above actual values).*
[3] *Simulation under conditions of 50% sparse weight and 90% inter-frame similarity.*
[4] *Simulation under 76.79% sparsity level of HARDVS (the higher the better).*
[5] *Simulation under 13.64% firing rate of HARDVS (the lower the better).*

RDCIM architecture, contributing to 53.4% of the performance improvement even at a 50% sparsity level.

C. Comparison With Previous Works

As shown in Table II, the RDCIM fusion accelerator achieves the peak performance of 42.02 TOPS/W for DFC under the sparsity level of HARDVS [18]. In SNN mode, the RDCIM fusion accelerator attains the energy consumption of 0.02 pJ/SOP. For the ResNet-18 model, the system efficiency is 27.41 TOPS/W in DFC and 0.03 pJ/SOP in SNN. Compared with previous works, our RDCIM architecture can provide remarkable energy efficiency when applied to the highly sparse fusion neural network.

A thorough comparison with state-of-the-art works is detailed in Table II. Benefiting from three optimization techniques, the RDCIM fusion accelerator accomplishes a significant area reduction of over 0.08 mm^2. These optimizations boost the area efficiency to 0.72 TOPS/mm^2 for DFC, representing a 3.4-fold improvement over the work [14]. These findings highlight the superior area efficiency of our RDCIM architecture.

V. CONCLUSION

In this brief, we introduce a weight-compressed RDCIM fusion accelerator with novel bitwise in-memory Booth multiplication, enabling efficient fusion computations for both DFC and SNN. A fine-grained 2:4 structured sparsity method prunes half of the weights without any accuracy loss, saving over 40% of storage resources. The concurrent support for three sparse optimization strategies facilitates the low-power implementation of the fusion accelerator. Implemented by TSMC 28nm node, our proposed RDCIM architecture achieves the peak energy efficiency of 42.02 TOPS/W for DFC and 0.02 pJ/SOP for SNN.

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